

BRAC University
CSE340: Computer Architecture
Quiz#2

Duration : 20 Minutes

Total Marks : 20 Set: Clock Cycle

ID:

Name:

Sec:

1. Load the number **0xBACAD1E0** in the register x22. [2]
2. Find the machine code of only the selected(bold) line:

```
addi x5, x0, 5
addi x6, x0, 0
loop:
add x6, x6, x5
addi x5, x5, -1
bne x5, x0, loop
addi x10, x6, 0
addi x17, x0, 10
```

The starting address of the first instruction is 0x5000. Use X for the fields that are not given. [3]

3. Consider the following code:

```
int kajKormo(int a, int b, int c) {
    int d = c >> b;
    a = d | c;
    return a;
}
```

```
int p = 2, q = 3, r = 4;
s[12] = kajKormo(p, q, r);
s[13] = s[12] * 9
```

Construct the equivalent RISC-V code of the above-mentioned C code. Base address of array s is in register X20; Consider the array as a word array. Variables p, q and r are stored in x21, x22 and x23 respectively. [15]

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Duration : 20 Minutes

Total Marks : 20 Set: Double Word

ID:

Name:

Sec:

1. Load the number **0xABACA1F0** in the register x22. [2]
2. Find the machine code of only the selected(bold) line:

```
addi x7, x0, 5
addi x8, x0, 0
loop:
add x8, x8, x7
addi x7, x7, -1
bne x7, x0, loop
addi x11, x8, 0
addi x18, x0, 10
```

The starting address of the first instruction is 0x8000. Use X for the fields that are not given. [3]

3. Consider the following code:

```
int kamKaj(int p, int q, int r) {
    int s = r >> q;
    p = s | r;
    return p;
}
```

```
int a = 2, b = 3, c = 4;
d [12] = kamKaj(a, b, c);
d [13] = d [12] * 7
```

Construct the equivalent RISC-V code of the above-mentioned C code. Base address of array d is in register X18; Consider the array as a word array. Variables a, b and c are stored in x19, x20 and x21 respectively. [15]